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MILITARY SPECIFICATION

BUFFING AND POLISHING COMPOUNDS

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers buffing and polishing compounds used for metal finishing.

1.2 Classification.- The buffing and polishing compounds shall be of the following types and grades, as specified (see 6.2):

Type I - Rouge cake
Type II - Lime cake
Type III - Chrome cake
  Grade A - Fast cutting and coloring
  Grade B - Highly finished surface
Type IV - Emery cake

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

P-C-535	Cleaning Compound, Plasters' Electrocleaning, for Steel
RR-S-366	Sieves, Standard for Testing Purposes

FSC 5350

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Military

MIL-P-3816	Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The ingredient materials used in the manufacture of the buffing and polishing compounds shall be of high quality, suitable for the purpose intended, and when finally compounded, shall conform to this specification (see 4.4.2).

3.1.1 Type I - rouge cake.- The type I buffing and polishing compound shall contain red rouge (Fe_2O_3) (see 4.4.2.1).

3.1.2 Type II - lime cake.- The type II buffing and polishing compound shall contain lime (CaO) (see 4.4.2.1).

3.1.3 Type III - chrome cake.- The type III buffing and polishing compound shall contain synthetic aluminum oxide (Al_2O_3) and chromium oxide (Cr_2O_3) (see 4.4.2.1).

3.1.4 Type IV - emery cake.- The type IV buffing and polishing compound shall contain emery (Al_2O_3, Fe_3O_4, and Fe_2O_3) and synthetic silicon carbide (SiC), unless otherwise specified (see 4.4.2.1).

3.1.4.1 Abrasive particle size.- The particle size of the abrasive material used in type IV buffing and polishing compound shall be such that 100 percent of the particles shall pass through the No. 200 U. S. Standard sieve of Specification RR-S-366, and not less than 95 percent by weight of the particles shall pass through the No. 230 U. S. Standard sieve (see 4.4.2.2).

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3.2 Form and put-up.- The types I, II, and III buffing and polishing compounds shall be furnished in either stick or cake form. The type IV buffing and polishing compound shall be furnished in cake form (see 4.4.2).

3.2.1 Weight per unit.- Each stick or cake of types I, II, and III buffing and polishing compound shall have a minimum weight of 1 pound and a maximum weight of 3 pounds. Each cake of type IV buffing and polishing compound shall weigh 3 ±1/2 pounds (see 4.4.1).

3.2.2 Stick package.- When furnished in stick form, the buffing and polishing compound shall be encased in a paper carton which can be peeled to expose the compound for use (see 4.4.1).

3.3 Performance.- When tested as specified in 4.4.2.3, the compounds may leave on the processed work a deposit of binder or residue which shall be readily removable with electroplaters' cleaning compound conforming to Specification P-C-535. Each type and grade shall produce the following results with efficient utilization of time, effort, and material.

3.3.1 Type I - rouge cake.- The type I buffing and polishing compound shall produce a highly lustrous finish and color on gold, silver, brass, and gold- and silver-plated surfaces (see 4.4.2.3).

3.3.2 Type II - lime cake.- The type II buffing and polishing compound shall produce a highly lustrous finish free from cast or discoloration on copper, brass, nickel, and nickel-alloy surfaces (see 4.4.2.3).

3.3.3 Type III - chrome cake.-

3.3.3.1 Grade A.- The type III, grade A buffing and polishing compound shall enable fast cutting and coloring, and satisfactory coloring work where shape or racking causes burned or gray spots on hard chromium plate, decorative chromium, steel, and stainless steel (see 4.4.2.3).

3.3.3.2 Grade B.- The type III, grade B buffing and polishing compound shall produce a highly polished finish on hard chromium plate, decorative chromium, steel, and stainless steel surfaces (see 4.4.2.3).

3.3.4 Type IV - emery cake.- The type IV buffing and polishing compound shall enable satisfactory abrasive cutting and polishing of metals (see 4.4.2.3).

3.4 Effects.- When used in accordance with the manufacturer's instructions, the buffing and polishing compounds shall not be corrosive to the processed work and shall not be obnoxious or toxic to personnel (see 4.4.2.3).

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3.5 Workmanship.- The buffing and polishing compound ingredients shall be intimately assembled and processed in a manner that will produce the high quality materials necessary to conform to this specification. (see 4.4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance inspection.- Acceptance inspection shall consist of examination and tests of random samples to assure conformance to this specification.

4.3 Sampling.-

4.3.1 Size of lot.- For the purpose of sampling, a lot of buffing and polishing compound shall consist of a manufacturer's batch. If the material cannot be identified by batch, a lot shall consist of all the buffing and polishing compound of one type or grade (as applicable) offered for delivery at one time. A batch is defined as the end product of all the raw materials mixed or blended in a single operation. The unit of inspection shall be one stick or cake.

4.3.2 Samples for examination of filled containers.- A random sample of filled containers of buffing and polishing compound shall be selected from each lot offered for acceptance under contract, in accordance with Standard MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective (see 4.4.1).

4.3.3 Samples for tests.- The inspector shall take at random one stick or cake of buffing and polishing compound from each lot. If more than one batch of buffing and polishing compound is included in the lot, one stick or cake shall be taken from each batch.

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4.4 Inspection methods.- Unless otherwise specified, inspection shall be in accordance with Standard MIL-STD-105. Unless otherwise specified, the physical and chemical values specified in section 3 apply to the average of the determinations made on the unit of product.

4.4.1 Examination of filled containers.- Each filled container selected in accordance with 4.3.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample container shall also be weighed to determine the amount of the contents. Any container in the sample, having one or more defects or under required fill, shall be rejected. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL-STD-105, the lot represented by the samples shall be rejected.

4.4.2 Examination and tests.- The ingredient materials, form and put-up, performance, effects, and workmanship requirements shall be examined and tested to assure conformance to 3.1, 3.2, 3.3, 3.4, and 3.5.

4.4.2.1 The composition of the chemical compounds specified in 3.1 shall be verfied by qualitative analysis methods.

4.4.2.2 Abrasive particle size (type IV only).- The following procedure shall be performed: Weigh 50 grams of the sample, transfer to a 300- or 400-milliliter beaker, while stirring, add about 150 milliliters of acetone; let the mineral matter settle, and pour off the supernatant liquid. Repeat three or four times until the fat or grease has been completely extracted. Dry the mineral matter on a steam bath in a beaker. Weigh 20 grams of the mineral matter, secured as described above, transfer to a nest of 200- and 230-mesh, United States Standard sieves and determine, by vibrating and oscillating the set, whether all of the material will pass through the 200-mesh sieve. If all of the material passes through the 200-mesh sieve, remove this sieve from the nest and wash the material remaining on the 230-mesh sieve under a stream of water; dry, weigh, and determine the percentage by weight of material remaining on the 230-mesh sieve (see 3.1.4.1).

4.4.2.3 Unbuffed and unpolished surfaces of the appropriate metals shall be utilized to determine satisfactory performance of the buffing and polishing compounds when used in accordance with the manufacturer's instructions (see 3.3). Effects of the buffing and polishing compounds shall be noted during and after the performance determinations (see 3.4).

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- The buffing and polishing compounds shall be packaged, packed, and marked in accordance with Specification MIL-P-3816 for specialties. Both the level of packaging and the level of packing employed shall be as specified by the procuring activity (see 6.2).

5.1.1 In addition to the marking specified in 5.1, individual containers shall be marked with the manufacturer's recommendations for use of the buffing and polishing compound and directions for obtaining optimum results in application.

6. NOTES

6.1 Intended use.- The buffing and polishing compounds covered by this specification are intended for use for metal finishing by buffing and polishing. The differences in use among the types and grades are as follows:

Type I - rouge cake for use where the highest luster, finish, and color is required for gold, silver, brass, and gold- and silver-plated articles.

Type II - lime cake for buffing copper, brass, nickel, and alloys of high nickel content to obtain a highly lustrous finish free from cast or discoloration.

Type III - chrome cake for cutting, polishing, buffing, and coloring hard chromium plate, decorative chromium, and stainless and other steels.

Grade A - for fast cutting and coloring or for coloring work where shape or racking causes burned or gray spots.

Grade B - for use where an extremely highly polished finish is desired.

Type IV - emery cake for use in abrasive cutting and polishing of metals.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, and grade (if applicable) (see 1.2).
(c) Quantity in pounds required.
(d) Levels of packaging and packing (see 5.1).
(e) Instructions regarding inspection records (see 4.1).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as specified in Specification MIL-P-3816, may be exercised in the preparation of buffing and polishing compounds for delivery.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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